



D2 Protein synthesis

D · Continuity and change · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Gene expression begins with transcription in the nucleus, where RNA polymerase uses a DNA template to synthesise a pre-mRNA molecule; in eukaryotes this is processed by splicing out introns, joining exons, and adding a 5' cap and poly-A tail to form mature mRNA. In translation, ribosomes read the mRNA in triplet codons; the near-universal, degenerate genetic code specifies which amino acid (or stop signal) each codon represents, and tRNA molecules, each carrying a specific amino acid, pair their anticodon with the matching mRNA codon so the ribosome can catalyse formation of peptide bonds, building a polypeptide that folds into a functional protein.

Worked example: follow the reasoning

Translating the mRNA sequence 5'-AUG GCU UUU UGA-3' into a polypeptide using the genetic code. Step 1: divide the sequence into codons (triplets) read from the 5' end: AUG | GCU | UUU | UGA. Step 2: look up each codon: AUG codes for methionine (also the start codon), GCU codes for alanine, UUU codes for phenylalanine, and UGA is a stop codon (does not code for an amino acid). Step 3: assemble the amino acid sequence in order, stopping translation at the stop codon. Conclusion: the resulting polypeptide is methionine-alanine-phenylalanine (Met-Ala-Phe), and the ribosome then releases the completed chain at the stop codon.

Quick check · multiple choice

What is the main function of a tRNA molecule during translation?

- A It catalyses peptide bond formation on its own
- B It carries a specific amino acid and pairs its anticodon with the complementary mRNA codon
- C It forms the structural core of the ribosome
- D It transports mature mRNA out of the nucleus

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A gene's DNA template strand includes the triplet 3'-TTC-5', which is transcribed and translated as part of a larger polypeptide. A point mutation changes this triplet to 3'-TTA-5'. [7]
 - (a) Determine the original mRNA codon and the mutated mRNA codon produced from these template triplets. [2]
 - (b) Using the genetic code (AAG = lysine, AAU = asparagine), state the amino acids specified by the original and mutated codons, and name this type of point mutation. [2]
 - (c) Explain why some point mutations have little or no effect on the final protein's function, while others can be highly damaging. [3]

Final answer: _____



Answers and explanation

Question 1 [7]

- (a) mRNA is complementary and antiparallel to the DNA template. Original template 3'-TTC-5' gives mRNA codon 5'-AAG-3'. Mutated template 3'-TTA-5' gives mRNA codon 5'-AAU-3'.
- (b) Original codon AAG codes for lysine; mutated codon AAU codes for asparagine. Since the codon still codes for an amino acid (just a different one), this is a missense mutation.
- (c) Effects depend on the chemical properties of the original versus new amino acid and their position in the protein: if the new amino acid has similar chemical properties (e.g. similar polarity/charge) and is not in a critical region (e.g. not in the active site of an enzyme), the protein's tertiary structure and function may be largely unaffected (a 'silent' or tolerated substitution); however, if the new amino acid has very different properties or occurs in a functionally critical region (e.g. active site, a site forming key bonds for tertiary structure), it can disrupt folding or function, sometimes severely, as seen in genetic diseases caused by single amino acid changes.

Quick check answer

B. Each tRNA is charged with one specific amino acid and recognises the corresponding mRNA codon via complementary base pairing between its anticodon and the codon, ensuring amino acids are added in the correct order.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

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